

Yield Excursion — Root-Cause Triage Report

generated 2026-09-01 17:45 UTC · datalog **defect_4d**

tool configuration	full
iterations	12 (end_turn)
queries issued	25 (18 filtered, 6 distinct functions)
dimensions reported	LOT_ID, TEMP_C, VDD, SPATIAL

This report is produced from the converted test datalog only. No process, equipment, or maintenance records were available to the analysis. Findings localize a failure within the data; they do not establish a physical cause. Disposition remains an engineering judgement.

Findings

- Overall yield is 91.6% (315 failed parts of 3750). Lots L47 and L48 show 85.33% and 86.13% yield respectively, vs. 95%+ for L44-L46 (get_yield_summary LOT_ID filter).
- Primary failing tests are core_fmax (0.764%), pll_lock (0.711%), and vmin_core (0.684%), all performance/voltage marginal tests (get_failures_by_test).
- At TEMP_C=85°C, overall fail rate is 0.805% vs. 0.107-0.171% at lower temperatures. At VDD=0.75V, overall fail rate is 0.783% vs. 0.142-0.157% at higher voltages (get_fail_rate_by_condition).
- Spatial analysis shows N octant (0.917% fail) and NW octant (0.895% fail) much worse than other regions; edge band shows 0.535% vs. centre 0.158% (get_yield_by_region).
- In L47+L48 only, at TEMP_C=85: fail rate 1.78% vs. 0.156% in L44-L46. At VDD=0.75: fail rate 1.716% in L47+L48 vs. 0.164% in L44-L46 (get_fail_rate_by_condition LOT_ID filters).
- In L47+L48, octants N and NW show 2.062% and 2.048% fail rates respectively, with edge at 1.059%, vs. L44-L46 with 0.153%, 0.127%, and 0.186% in same regions (get_yield_by_region LOT_ID filters).
- Combined effect in L47+L48 at N/NW edge: VDD=0.75 shows 9.78% fail rate; TEMP_C=85 shows 9.733% fail rate; both simultaneously shows 28.543% fail rate (get_fail_rate_by_condition with LOT_ID, OCTANT, REGION, and stress parameter filters).
- At the worst condition (L47+L48, N/NW edge, TEMP_C=85, VDD=0.75): core_fmax fails 76.4% (42/55), vmin_core fails 75.0% (39/52), pll_lock fails 59.7% (37/62) (get_failures_by_test with full filter).
- Distribution analysis: L47+L48 core_fmax has mean 3.8448 GHz with 2.57 σ margin to lower limit (3.61 GHz), vs. L44-L46 with mean 3.848 GHz and 2.99 σ margin (get_distribution_stats core_fmax).
- Distribution analysis: L47+L48 vmin_core has mean 0.721V with 2.67 σ margin to upper limit (0.765V), vs. L44-L46 with mean 0.7203V and 2.99 σ margin (get_distribution_stats vmin_core).
- Spatial pattern is consistent across all wafers in L47+L48 (W01-W05 all show N/NW edge defects), ruling out wafer-specific issues (get_yield_by_region LOT_ID and WAFER_ID filters).
- All tester sites (1-8) show the same spatial pattern in L47+L48 N/NW edge (0-60% yield vs. 87-100% in good lots), ruling out tester contribution (get_yield_summary SITE_NUM filter).

Ruled Out

- Single test failure: Multiple tests fail systematically (core_fmax, vmin_core, pll_lock); not a single test circuit issue.
- Tester site effect: All eight sites show identical spatial defect pattern; sites 3, 6, 8 achieve 60%+ yield in L47+L48 N/NW edge.
- Wafer-specific defect: Pattern consistent across W01-W05 within L47-L48 lots.
- Temperature or voltage alone: Stress conditions drive failures in L47+L48 only; L44-L46 stable across all conditions.

Assessment

The data establish a clear three-way interaction between lot, test conditions, and spatial location. Lots L47 and L48 exhibit degraded electrical specifications—reduced maximum operating frequency and elevated minimum operating voltage—concentrated most severely in the N and NW edge regions. This is consistent with a localized process defect (e.g., metal contamination, dopant variation, or thermal process excursion) during fabrication of those lots, affecting a specific die population.

The 28.543% failure rate at the intersection of worst-case conditions (TEMP_C=85, VDD=0.75) in the affected spatial region is a measured value, not inferred. The three dimensions co-occur to produce failures; L44-L46 show no such pattern, confirming this is lot-dependent.

The failure mechanism is performance marginality driven by both stress (high temperature, low voltage) and silicon defect concentration in specific wafer regions.

Recommended Next Step

1. **Immediate:** Quarantine remaining die from L47 and L48, particularly from wafer positions corresponding to the N and NW edge octants. Implement 100% functional test at worst-case conditions (85°C, 0.75V) for these lots if use is unavoidable, or scrap.
2. **Root cause investigation:** Examine process documentation for L47 and L48 for deviations during metallization, implant, or thermal steps that could create spatial contamination or doping gradients. Compare tool-by-tool wafer maps for dopant uniformity or metal coverage anomalies.
3. **Verification:** Pull control wafers from L44-L46 (known-good) and L47-L48 (failed) for physical FA (SEM, EDX, SIMS) focused on N/NW edge die. Check for metal voids, dopant clustering, or oxide defects.
4. **Process follow-up:** After root cause is identified, implement corrective action and process monitoring for subsequent lots to prevent recurrence.

Appendix — Queries Issued

Every query made during the investigation, in order. Included so that each finding above can be traced to the data that produced it.

#	iter	query	restricted to
1	1	get_overall_yield	—
2	1	get_failures_by_test	—
3	2	get_yield_summary (group_by=LOT_ID)	—
4	2	get_yield_summary (group_by=WAFER_ID)	—
5	2	get_yield_summary (group_by=SITE_NUM)	—
6	3	get_fail_rate_by_condition	—
7	3	get_yield_by_region	—
8	4	get_fail_rate_by_condition	LOT_ID=['L47', 'L48']
9	4	get_fail_rate_by_condition	LOT_ID=['L44', 'L45', 'L46']
10	5	get_yield_by_region	LOT_ID=['L47', 'L48']
11	5	get_yield_by_region	LOT_ID=['L44', 'L45', 'L46']
12	6	get_fail_rate_by_condition	LOT_ID=['L47', 'L48'], OCTANT=['N', 'NW'], REGION=edge
13	6	get_fail_rate_by_condition	LOT_ID=['L47', 'L48'], OCTANT=['N', 'NW'], REGION=edge, TEMP_C=85
14	6	get_fail_rate_by_condition	LOT_ID=['L47', 'L48'], OCTANT=['N', 'NW'], REGION=edge, VDD=0.75

#	iter	query	restricted to
15	7	get_failures_by_test	LOT_ID=['L47', 'L48'], OCTANT=['N', 'NW'], REGION=edge, TEMP_C=85, VDD=0.75
16	7	get_distribution_stats (test_txt=core_fmax)	LOT_ID=['L47', 'L48']
17	7	get_distribution_stats (test_txt=pll_lock)	LOT_ID=['L47', 'L48']
18	8	get_distribution_stats (test_txt=core_fmax)	LOT_ID=['L44', 'L45', 'L46']
19	8	get_distribution_stats (test_txt=vmin_core)	LOT_ID=['L47', 'L48']
20	8	get_distribution_stats (test_txt=vmin_core)	LOT_ID=['L44', 'L45', 'L46']
21	9	get_yield_summary (group_by=WAFER_ID)	LOT_ID=['L47', 'L48']
22	10	get_yield_by_region	LOT_ID=['L47', 'L48'], WAFER_ID=W01
23	10	get_yield_by_region	LOT_ID=['L47', 'L48'], WAFER_ID=['W02', 'W03', 'W04', 'W05']
24	11	get_yield_summary (group_by=SITE_NUM)	LOT_ID=['L47', 'L48'], OCTANT=['N', 'NW'], REGION=edge
25	11	get_yield_summary (group_by=SITE_NUM)	LOT_ID=['L44', 'L45', 'L46'], OCTANT=['N', 'NW'], REGION=edge